

FINANCIAL DISCLOSURE
AACPDM 80th Annual Meeting
September 27-30, 2026



Speaker Name: Katharina Quinlan

1. Disclosure of Relevant Financial Relationships

I have the following financial relationships to disclose:

Grant/Research support from:

National Institutes of Neurological Disease and Stroke

R01 NS104436

R01 NS132728

R01 NS135580

Employee of:

University of Rhode Island College of Pharmacy and
George and Anne Ryan Institute for Neuroscience

2. Disclosure of Off-Label and/or investigative uses:

I will discuss the following off label use and/or investigational use in my presentation:

Zolmitriptan (Zomig®)



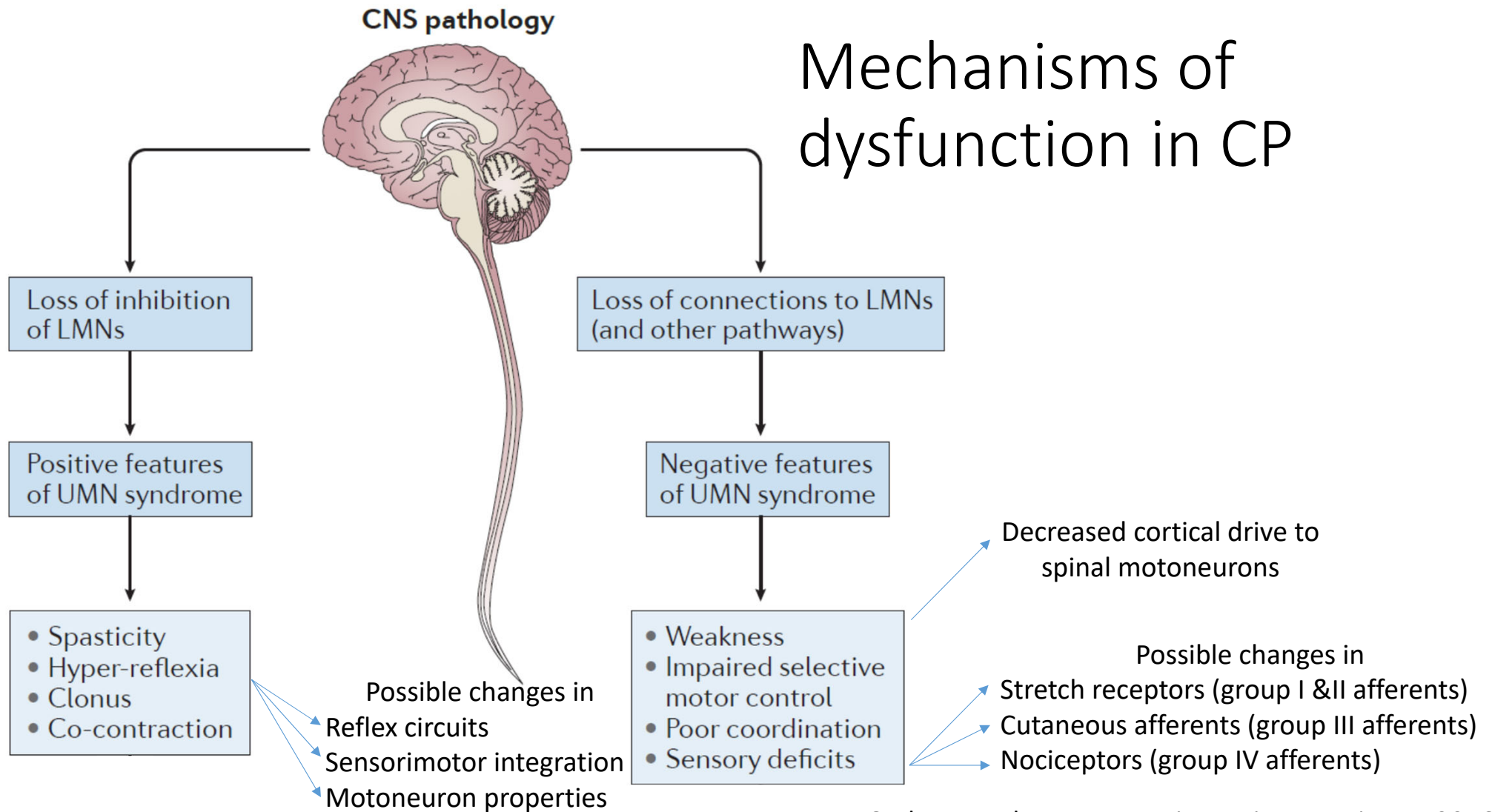
PHILADELPHIA
— SEPTEMBER 27-30, 2026 —
Rooted in Brotherly Love & Lifelong Community

ANNUAL
MEETING

Spinal Cord Changes in CP: The Missing Link?

How brain injury affects spinal motoneurons

Mechanisms of dysfunction in CP



From Graham et al., *Nature Reviews Disease Primers*, 2016

Major goals for spinal cord research

Find novel targets
for treatment of spasticity
by targeting synapses
in the reflex circuit



Hyperreflexia in CP

Spasticity = Velocity-dependent change in stiffness of a limb / muscle

- ✓ Larger amplitude of H reflex
- ✓ Less depression of the H reflex during repeated stimuli
 - rate dependent depression
 - post activation depression
- ✓ More spread of the H reflex (reflex irradiation)

Cortical control of spinal reflexes

Corticospinal tract (CST)

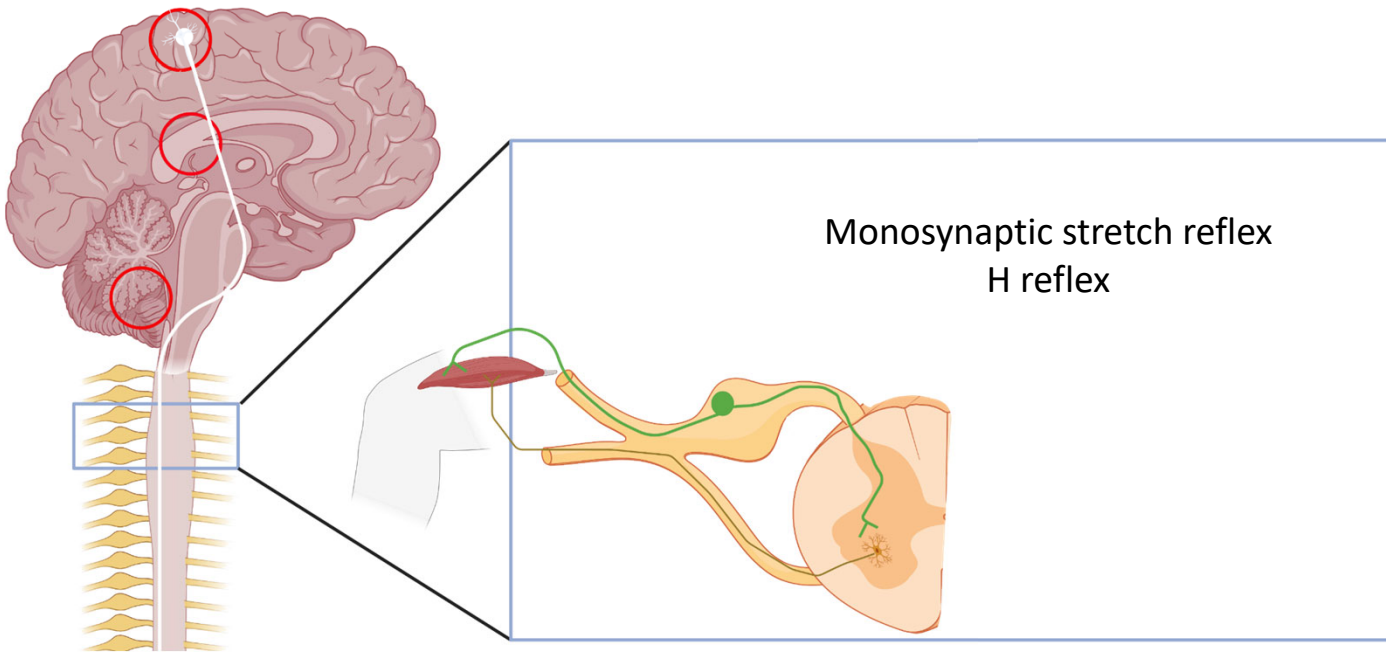


Image created with
Biorender

Reflexes altered by:

CST

GABAergic (GAD2) interneurons

Glutamatergic (Vglut3) interneurons

Primary afferent depolarization

Introduction to the rabbit model



Sham control



Prenatal hypoxia-ischemia (HI) injury

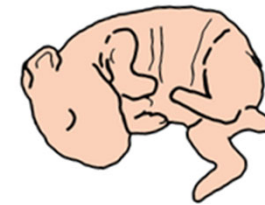
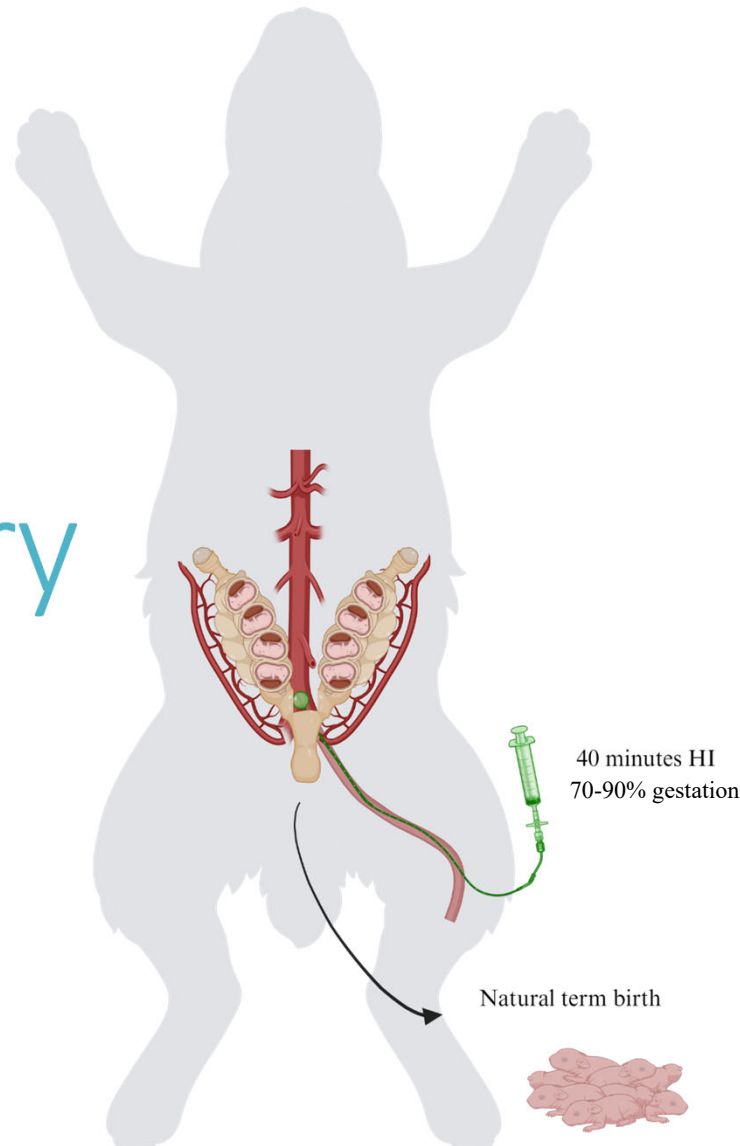


Hypoxia-ischemia (HI)

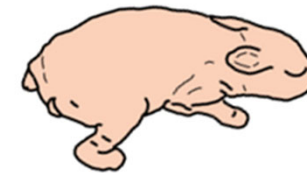
Prenatal hypoxia-ischemia (HI) surgery



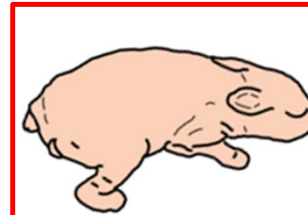
Developed by
Professor Sidhartha Tan, MD
Derrick et al., (2004) *J Neurosci*



Motor-affected
HI (hypertonia)



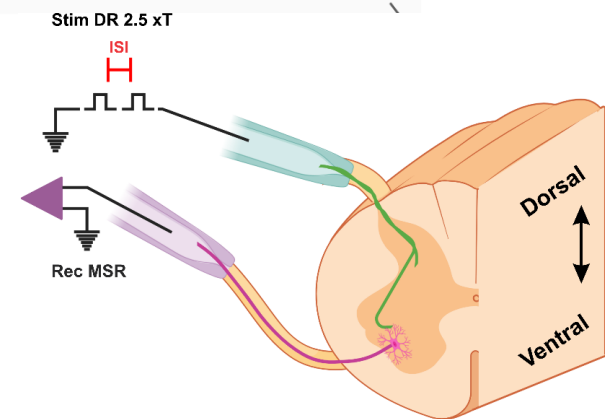
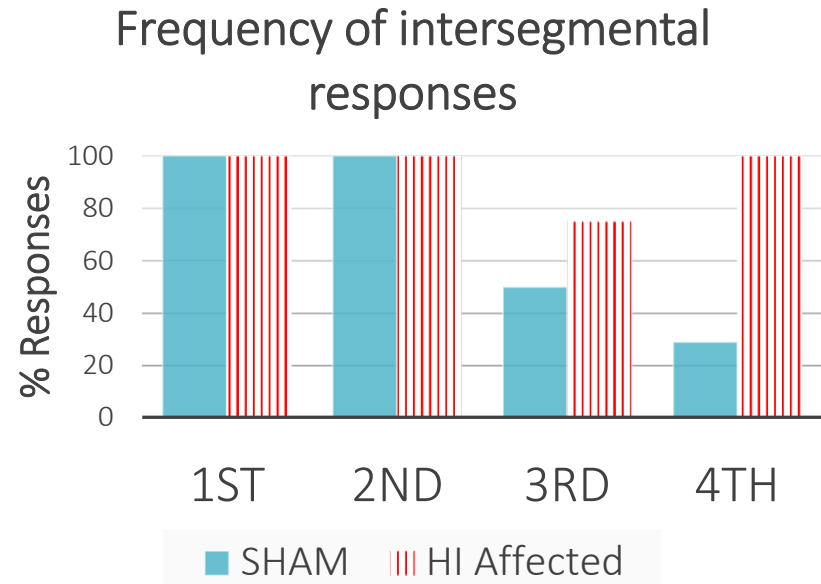
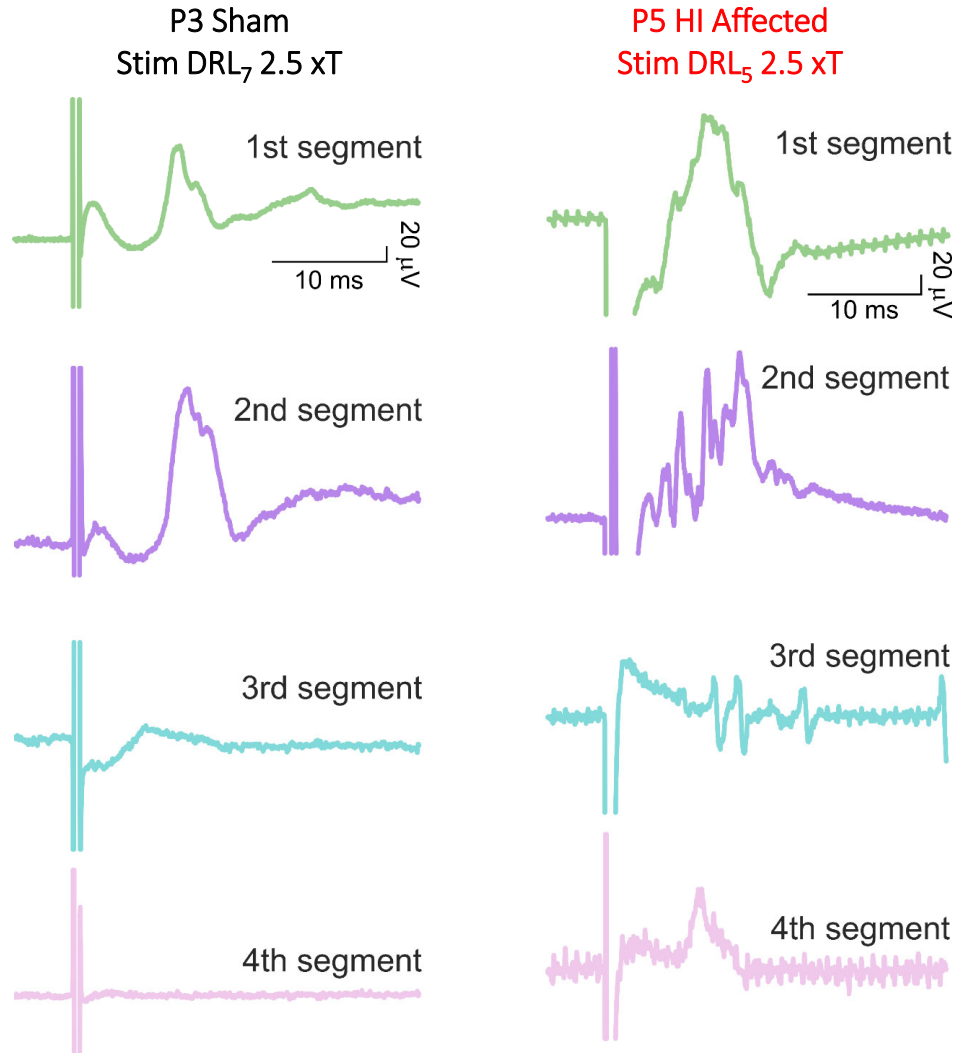
Motor-
unaffected HI



Shams: surgery
without HI

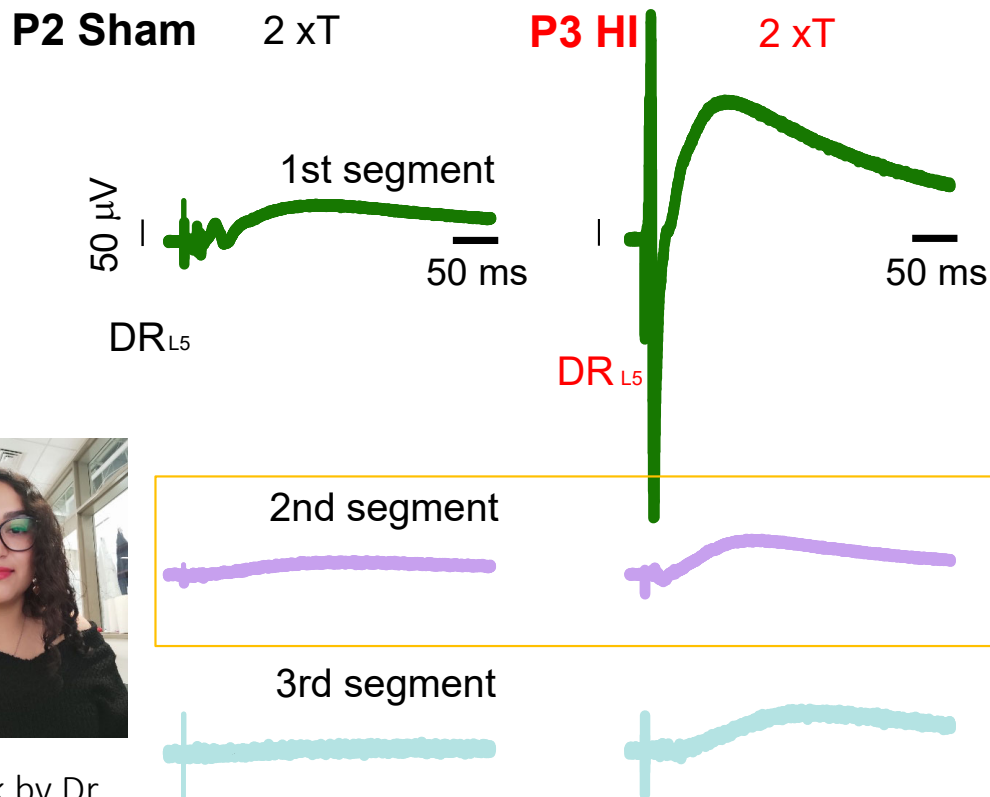
Surgical controls: separate surgery

Reflex irradiation is increased in HI rabbits



Work by Dr
Mena Avila

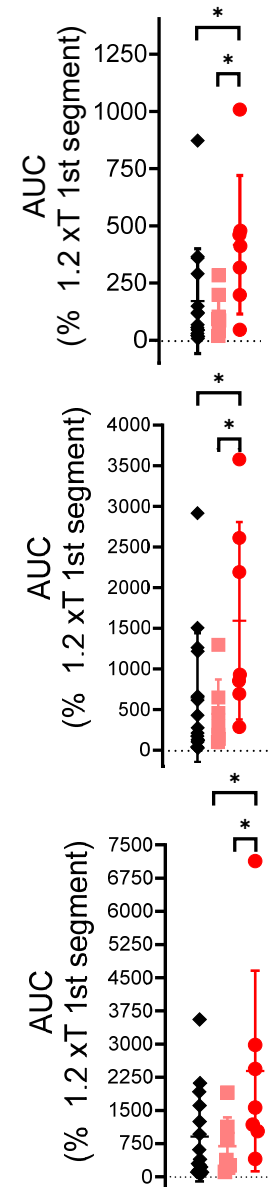
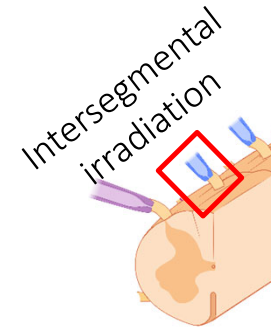
Primary afferent depolarization is larger in HI affected rabbits



2nd segment,
normalized to 1.2xT

- ◆ Sham
- HI Unaffected
- HI Affected

Two-way ANOVA Tukey's Test
* $p < 0.05$



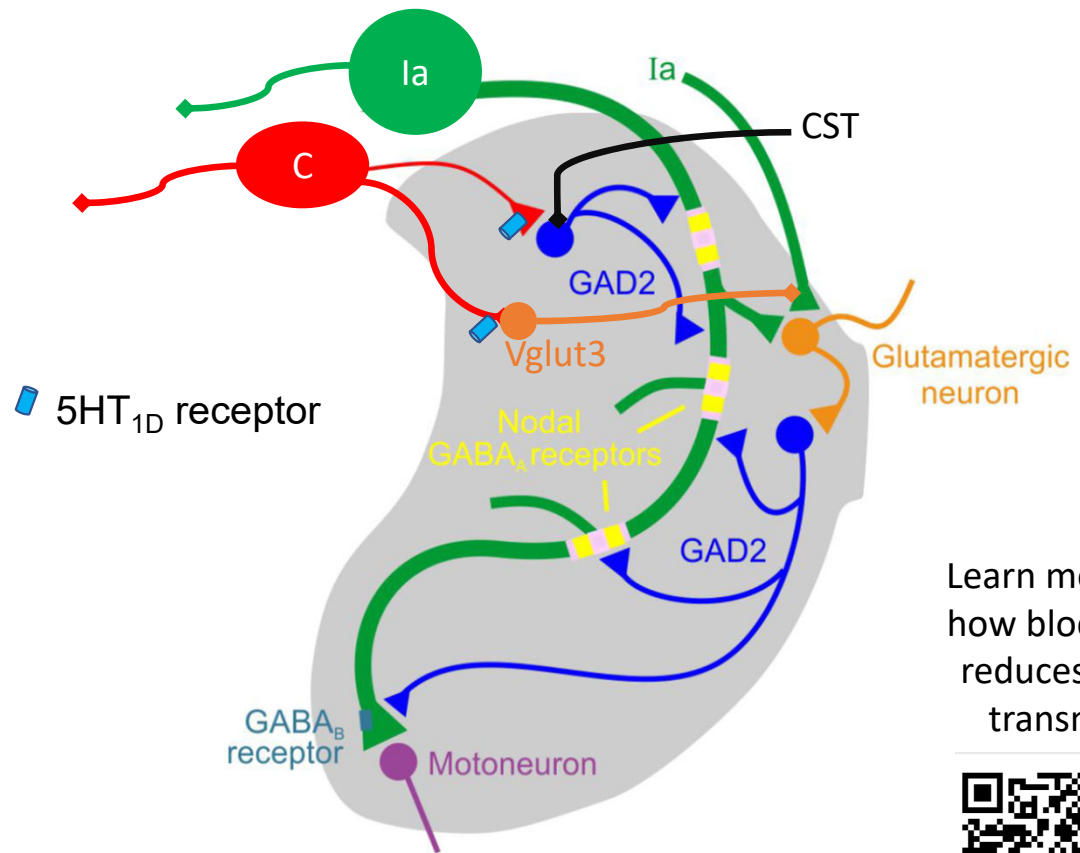
2 xT (group I/II, β) 5 xT (group III, δ) 10 xT (group IV, γ)



Work by Dr
Mena Avila

Primary afferent depolarization circuit

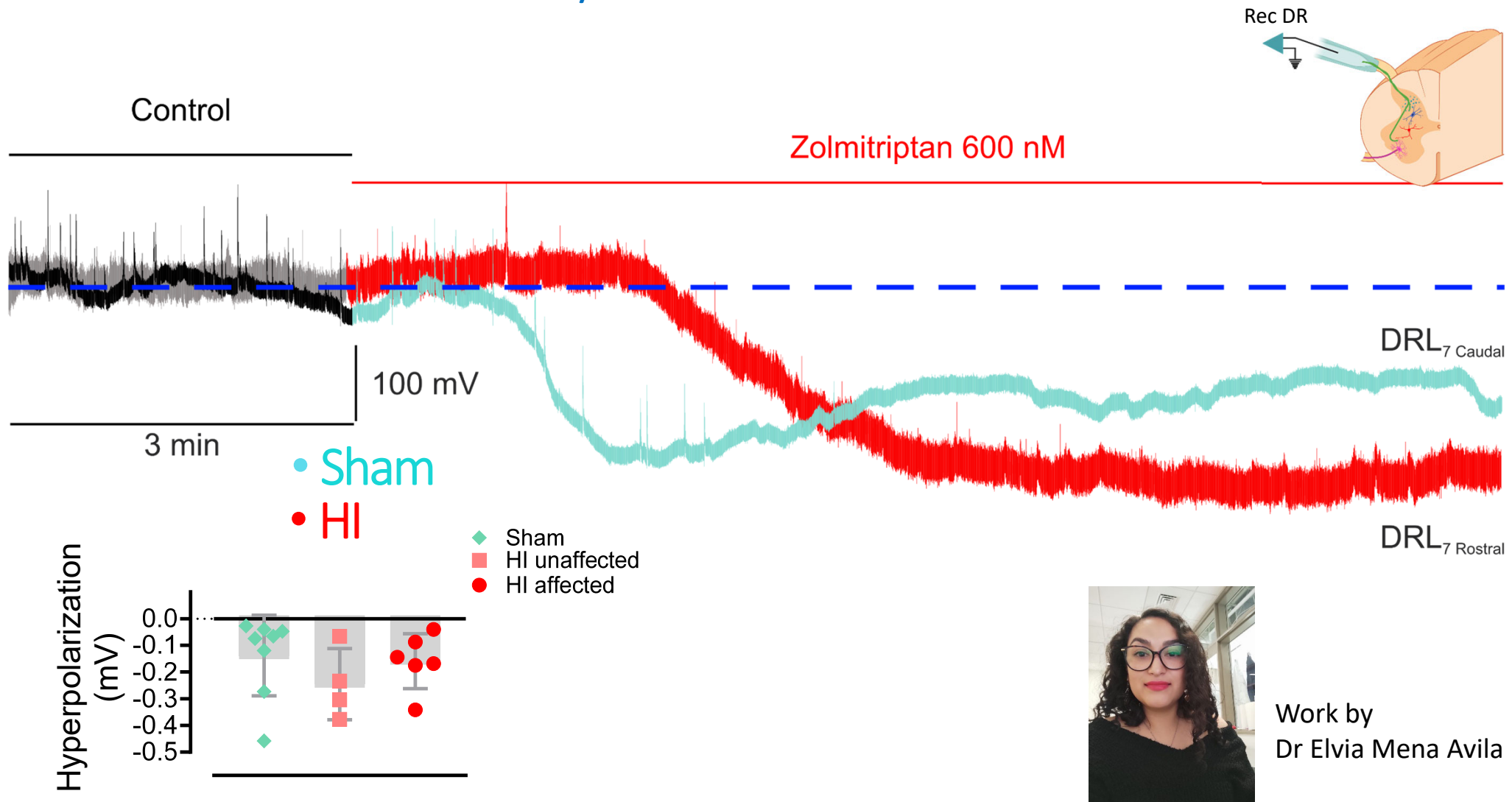
- There is more primary afferent depolarization (PAD) in HI spinal cords
- It may be possible to reduce PAD and spasticity with neuromodulators like serotonin



Learn more about how blocking PAD reduces sensory transmission

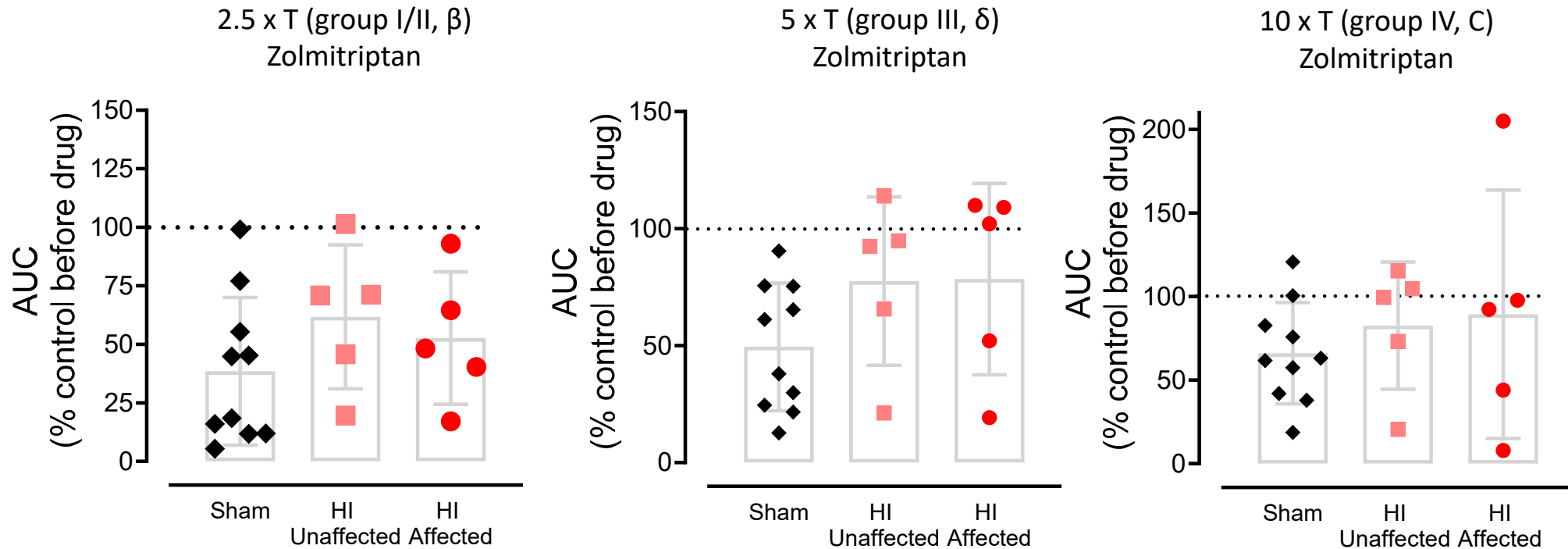


Activation of 5HT_{1B/D} receptors hyperpolarizes afferents



Work by
Dr Elvia Mena Avila

5-HT_{1B/D} agonist zolmitriptan reduces primary afferent depolarization

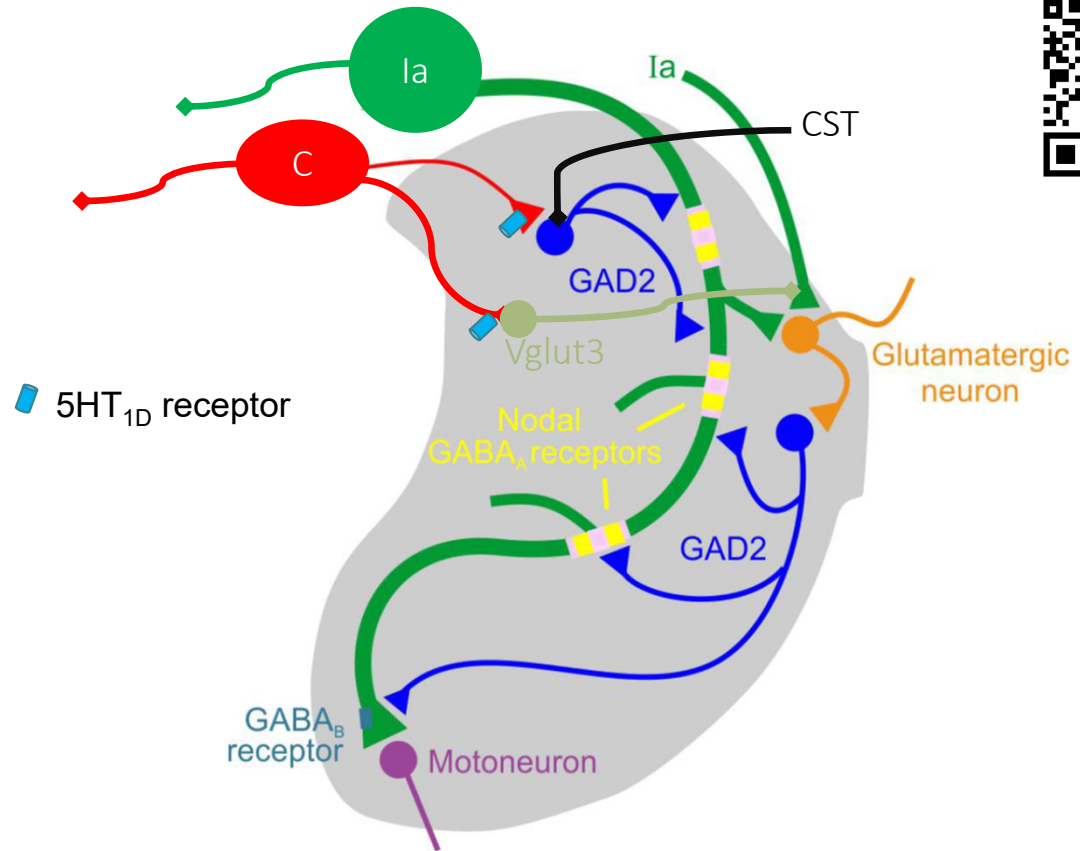


Work by
Dr Elvia Mena Avila



Take home messages:

- Brain injuries result in weaker corticospinal tract (CST)
- This results in more chronically activated primary afferent depolarization (PAD)
- It may be possible to dampen excitability of afferents and reduce PAD and associated spasticity
- Spinal circuits provide emerging targets for clinical care



VISIT our posters Monday morning: SP46, SP156, Tues morning SP141

Thanks & Acknowledgements

The Quinlan lab team



Landon Genry, Dr Elvia Mena Avila,
Brendan Moline, Dr Emily Reedich,
Lisa Dowaliby, Cailyn Doyle, Paola
Sustache-Rivera, Jenna Lees,
Cassandra Kramer and Tracy Santos



CP Community partners

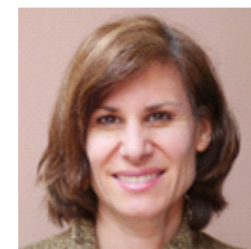


Ross Bostick, Ashleigh Nightengale,
Ann Adams

Collaborators:



Marin Manuel
U of Rhode Island



Monica Gorassini
University of Alberta



Dave Bennett
University of Alberta



Megan Detloff
Drexel
University



Ben Katholi



Deb Gaebler-Spira

Shirley Ryan Abilitylab